

Mr./Ms. Daniel Conde-Torres

Title of the thesis: Antimicrobial Peptides at Membrane Interfaces: Classical and Quantum Approaches to Environment-Conditioned Peptide Design

As established in Article 37.2.c) of the Doctoral Studies Regulations of the USC:

*c) The doctoral candidate must also submit:*

*i. A sworn statement declaring that the contributions included in the thesis are not part of any other thesis and that they are original and the candidate's own work.*

*ii. A report from the thesis supervisor expressing their agreement with the submission of the thesis as a compilation of publications, justifying that the contributions included in the compilation align with the candidate's research plan, and stating that the candidate's contribution is substantial and relevant<sup>1</sup>.*

*iii. A written waiver from non-PhD co-authors, renouncing the inclusion of the scientific contributions as part of their own doctoral theses.*

**I hereby declare responsibly that:**

I am submitting the thesis in the format of a thesis by compilation of publications.

The contributions included in the thesis are not part of any other thesis and are original and my own work.

In the event that there are other non-PhD co-authors in these contributions, I enclose as attached documentation their written consent to waive the submission of the scientific contributions as part of their doctoral theses.

Electronic signature: Daniel Conde Torres

---

<sup>1</sup> This procedure will be considered completed once the thesis supervisors have issued a favorable report on the authorization request for the defense in the online system.

LIST OF SCIENTIFIC CONTRIBUTIONS FOR THIS MODALITY

|                                                                                                                                                                                                                                                                                                    |                                                                                       |                                     |                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
| <b>Title and full citation of the publication</b>                                                                                                                                                                                                                                                  |                                                                                       |                                     |                                     |
| <b>Unraveling lipid and inflammation interplay in cancer, aging and infection for novel theranostic approaches</b>                                                                                                                                                                                 |                                                                                       |                                     |                                     |
| Conde-Torres, D., <i>et al.</i> Unraveling lipid and inflammation interplay in cancer, aging and infection for novel theranostic approaches. <i>Frontiers in Immunology</i> <b>15</b> , 1320779 (2024). doi: <a href="https://doi.org/10.3389/fimmu.2024.1320779">10.3389/fimmu.2024.1320779</a> . |                                                                                       |                                     |                                     |
| <b>Authorship</b>                                                                                                                                                                                                                                                                                  | <input type="checkbox"/> I am the only author                                         |                                     |                                     |
|                                                                                                                                                                                                                                                                                                    | <input checked="" type="checkbox"/> The publication includes the following co-authors |                                     |                                     |
|                                                                                                                                                                                                                                                                                                    | <b>Name of the co-author<br/>(including author)</b>                                   | <b>Doctor</b>                       | <b>Non doctor</b>                   |
|                                                                                                                                                                                                                                                                                                    | Daniel Conde-Torres                                                                   | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
|                                                                                                                                                                                                                                                                                                    | Alexandre Blanco-González                                                             | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                    | Alejandro Seco-González                                                               | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                    | Fabián Suárez-Lestón                                                                  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                    | Alfonso Cabezón                                                                       | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                    | Paula Antelo-Riveiro                                                                  | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
|                                                                                                                                                                                                                                                                                                    | Ángel Piñeiro                                                                         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Rebeca García-Fandiño                                                                                                                                                                                                                                                                              | <input checked="" type="checkbox"/>                                                   | <input type="checkbox"/>            |                                     |

|                                                                                                                                                                                                                                                                                                                          |                                                                                       |                                     |                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
| <b>Title and full citation of the publication</b>                                                                                                                                                                                                                                                                        |                                                                                       |                                     |                                     |
| <b>Classical Simulations on Quantum Computers: Interface-Driven Peptide Folding on Simulated Membrane Surfaces</b>                                                                                                                                                                                                       |                                                                                       |                                     |                                     |
| Conde-Torres, D., <i>et al.</i> Classical Simulations on Quantum Computers: Interface-Driven Peptide Folding on Simulated Membrane Surfaces. <i>Computers in Biology and Medicine</i> <b>182</b> , 109157 (2024). doi: <a href="https://doi.org/10.1016/j.compbiomed.2024.109157">10.1016/j.compbiomed.2024.109157</a> . |                                                                                       |                                     |                                     |
| <b>Authorship</b>                                                                                                                                                                                                                                                                                                        | <input type="checkbox"/> I am the only author                                         |                                     |                                     |
|                                                                                                                                                                                                                                                                                                                          | <input checked="" type="checkbox"/> The publication includes the following co-authors |                                     |                                     |
|                                                                                                                                                                                                                                                                                                                          | <b>Name of the co-author<br/>(including author)</b>                                   | <b>Doctor</b>                       | <b>Non doctor</b>                   |
|                                                                                                                                                                                                                                                                                                                          | Daniel Conde-Torres                                                                   | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
|                                                                                                                                                                                                                                                                                                                          | Mariamo Mussa-Juane                                                                   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                                          | Daniel Faílde                                                                         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                                          | Andrés Gómez                                                                          | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                                          | Rebeca García-Fandiño                                                                 | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Ángel Piñeiro                                                                                                                                                                                                                                                                                                            | <input checked="" type="checkbox"/>                                                   | <input type="checkbox"/>            |                                     |

|                                                                                                                                                                                                                                                                                       |                                                                                       |                                     |                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
| <b>Title and full citation of the publication</b>                                                                                                                                                                                                                                     |                                                                                       |                                     |                                     |
| <b>Environment-Conditioned Design of <math>\alpha</math>-Helical Peptides</b>                                                                                                                                                                                                         |                                                                                       |                                     |                                     |
| Conde-Torres, D., García-Fandiño, R. & Piñeiro, Á. Environment-Conditioned Design of $\alpha$ -Helical Peptides. <i>Journal of Chemical Theory and Computation</i> <b>22</b> , 9776–9789 (2026). doi: <a href="https://doi.org/10.1021/acs.jctc.6c01278">10.1021/acs.jctc.6c01278</a> |                                                                                       |                                     |                                     |
| <b>Authorship</b>                                                                                                                                                                                                                                                                     | <input type="checkbox"/> I am the only author                                         |                                     |                                     |
|                                                                                                                                                                                                                                                                                       | <input checked="" type="checkbox"/> The publication includes the following co-authors |                                     |                                     |
|                                                                                                                                                                                                                                                                                       | <b>Name of the co-author<br/>(including author)</b>                                   | <b>Doctor</b>                       | <b>Non doctor</b>                   |
|                                                                                                                                                                                                                                                                                       | Daniel Conde-Torres                                                                   | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
|                                                                                                                                                                                                                                                                                       | Rebeca García-Fandiño                                                                 | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                       | Ángel Piñeiro                                                                         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |                                     |                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
| <b>Title and full citation of the publication</b><br><br><b>Unlocking the specificity of antimicrobial peptide interactions for membrane-targeted therapies</b><br><br>Conde-Torres, D., Calvelo, M., Rovira, C., Piñeiro, Á. & Garcia-Fandino, R. Unlocking the specificity of antimicrobial peptide interactions for membrane-targeted therapies. <i>Computational and Structural Biotechnology Journal</i> <b>25</b> , 61–74 (2024). doi: <a href="https://doi.org/10.1016/j.csbj.2024.04.022">10.1016/j.csbj.2024.04.022</a> . |                                                                                       |                                     |                                     |
| Authorship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <input type="checkbox"/> I am the only author                                         |                                     |                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <input checked="" type="checkbox"/> The publication includes the following co-authors |                                     |                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Name of the co-author (including author)</b>                                       | <b>Doctor</b>                       | <b>Non doctor</b>                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Daniel Conde-Torres                                                                   | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Martín Calvelo                                                                        | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Carme Rovira                                                                          | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Ángel Piñeiro                                                                         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Rebeca García-Fandiño                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <input checked="" type="checkbox"/>                                                   | <input type="checkbox"/>            |                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       |                                     |                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
| <b>Title and full citation of the publication</b><br><br><b>Mapping the quantum computing landscape: growth, collaboration, and thematic convergence</b><br><br>Conde-Torres, D., Seco-González, A., Piñeiro, Á. <i>et al.</i> Mapping the quantum computing landscape: growth, collaboration, and thematic convergence. <i>EPJ Quantum Technology</i> <b>13</b> , 7 (2026). doi: <a href="https://doi.org/10.1140/epjqt/s40507-026-00464-4">10.1140/epjqt/s40507-026-00464-4</a> |                                                                                       |                                     |                                     |
| Authorship                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <input type="checkbox"/> I am the only author                                         |                                     |                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <input checked="" type="checkbox"/> The publication includes the following co-authors |                                     |                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Name of the co-author (including author)</b>                                       | <b>Doctor</b>                       | <b>Non doctor</b>                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Daniel Conde-Torres                                                                   | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Alejandro Seco-González                                                               | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Ángel Piñeiro                                                                         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rebeca García-Fandiño                                                                 | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |

